

Date: 23/07/2025

TASK 1: Running Python script and various expression in an interactive Interpreter

a) perform Basic Mathematical Computations

AIM:

To write Python program that accepts two numerical inputs and performs addition, subtraction, multiplication, and division operations.

Algorithm:

1. Start the program
2. Accept two numerical inputs from the user.
3. Perform:
 - Addition
 - subtraction
 - multiplication
 - Division (if second number is not zero)
4. Display the results
5. End the program.

Program

```
num 1 = float(input("Enter first value : "))  
num 2 = float(input("Enter second value : "))  
Print("Addition:", num 1 + num 2)  
Print("Subtraction:", num 1 - num 2)  
Print("Multiplication:", num 1 * num 2)  
Print("Division:", num 1 / num 2)
```

OUTPUT

Enter first value : 100

Enter second value : 20

Addition : 120.0

Subtraction : 80.0

Multiplication : 2000.0

Division : 5.0

OR

VEL TECH - CSE	
EX NO.	
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VA VOCE (5)	
ORD (5)	
(20)	
WITH DATE	

RESULT:

The program successfully performed all arithmetic operation on the given inputs and displayed the results.

(b) Evaluate Relation Expressions

AIM:

To develop a python program that compares two numeric values using relational operators and display the result of each comparison.

ALGORITHM

1. Start the program.
2. Accept two numbers from the user.
3. Apply the following relational Operators:
 - Greater than ($>$)
 - Less than ($<$)
 - Equal to ($=$)
 - Not equal to ($\neq$)
 - Greater than or equal to ($\geq$)
 - Less than or equal to ($\leq$)
4. Display the results
5. End the program.

PROGRAM

```
a = float(input("Enter first score:"))
b = float(input("Enter second score:"))
print("a > b:", a > b)
print("a < b:", a < b)
print("a == b:", a == b)
print("a != b:", a != b)
print("a >= b:", a >= b)
print("a <= b:", a <= b)
```

OUT PUT

Enter first score: 85

Enter second score: 90

$a > b$: False

$a < b$: True

$a == b$: False

$a != b$: True

$a >= b$: False

$a <= b$: True

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VET TECH - CRR	
EX NO	
NAME	(B) EDWARDS
DATE	(C) 21/11/2019
TIME	(D) 10:00
DATE	(E) 01/12
TIME	(F) 10:00

VEL TECH - CSE	
EX NO.	
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VIVA VOCE (5)	
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	

RESULT:

The program correctly evaluated all the relational expressions between the two given inputs.

(c) check logical conditions across multiple inputs

AIM:

TO create a python program that uses logical operators (and, or, not) to evaluate conditions across these test scores.

ALGORITHM:

1. Start the program.
2. Accept three test scores from the user.
3. Use logical operators to evaluate:
 - If the candidate passed all tests (and)
 - If the candidate passed at least one test (or)
 - If the candidate failed all tests (not)
4. Display the results.
5. End the program.

PROGRAM

```
test 1 = int(input("Enter marks for Test 1:"))
test 2 = int(input("Enter mark for Test 2:"))
test 3 = int(input("Enter marks for Test 3:"))

print("passed all tests:", test 1 > 40 and test 2 > 40 and test 3 > 40)

print("passed at least one test:", test 1 > 40 or test 2 > 40 or test 3 > 40)

print("Failed all tests:", not (test 1 > 40 or test 2 > 40 or test 3 > 40))
```


OUTPUT

Enter marks for test 1 = 45

Enter marks for test 2 = 38

Enter marks for test 3 = 48

passed all the tests = False

passed at least one test = True

Failed all test : False



VEL TECH - CSE	
EX NO	1
PLR. NAME (P)	5
RESULT ANALYSIS (P)	5
VIVA SCORE	5
RECORD (P)	15
TOTAL (40)	15
SIGNATURE	

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RESULT:

The program effectively evaluated logical expressions and correctly identified pass/fail conditions based on test scores.